Proofs of T01 (contrast reduction and mixed-rate transfer) and T02 (dimension control and exclusion of strict supersets) for SMC-BVS v3

Lead: Claude Opus 5.5 (wave 3). Version 3 (final; v2 as reviewed in round 2, plus notation-only fixes).

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All notation, assumptions and statements are those of `STATEMENT.md`; we recall what is used. Throughout, $\Pi(\cdot \mid y)$ is the v3 posterior (on Ω, or jointly with the parameters), and Π_b is the contrast-model posterior at a *fixed* rate $b > 0$. No result outside this document is used, apart from elementary facts (Cauchy interlacing, the change-of-variables formula, Tonelli's theorem, Markov's and Chebyshev's inequalities, the moment generating function of a chi-square variable, and the fact that inversion reverses the Loewner order: if $0 \prec V_1 \preceq V_2$ then $V_1^{-1/2} V_2 V_1^{-1/2} \succeq I$, hence $V_1^{1/2} V_2^{-1} V_1^{1/2} \preceq I$, i.e. $V_2^{-1} \preceq V_1^{-1}$). In no step is ε or ε_c treated as $\mathcal{N}(0, \sigma_*^2 I)$ conditionally on s_y or on $\mathcal{B}_n$; every probability is computed from the unconditional Gaussian law of ε.

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1 Setting and conventions

Data. $n \geq 3$, $p \geq 1$, $C = I_n - \mathbf{1}\mathbf{1}^\top/n$, $X = CX^r D_x^{-1}$, $s_y^2 = \|Cy^r\|^2/(n-1) > 0$, $y = Cy^r/s_y$. $U \in \mathbb{R}^{n \times (n-1)}$ is a fixed (non-random) matrix with $U^\top U = I_{n-1}$ and $UU^\top = C$. Then $U^\top \mathbf{1} = 0$ (as $\|U^\top \mathbf{1}\|^2 = \mathbf{1}^\top C \mathbf{1} = 0$), $U^\top C = U^\top U U^\top = U^\top$, and $CX = X$. Put $y_c = U^\top y^r$ and $X_c = U^\top X$. Then

$$\begin{aligned} X &= UU^\top X = UX_c, \qquad y = UU^\top y^r/s_y = Uy_c/s_y, \\ X_c^\top X_c &= X^\top X, \qquad \|y_c\|^2 = (y^r)^\top C y^r = (n-1)s_y^2. \end{aligned} \tag{1}$$

Model quantities. For $S \in \Omega = \{S \subseteq [p] : |S| \le K\}$, $K = \min(p, \lfloor n/2 \rfloor)$, $k = |S|$: $G_S = X_S^\top X_S = (X_{c,S})^\top X_{c,S} = (n-1)R_S$, $H_S = (n+1)G_S + I_k$, $D_S = \det H_S / \det(G_S + I_k)$, $z_S = X_S^\top y$, $Q_S = y^\top y - n z_S^\top H_S^{-1} z_S$, $z_S^c = X_{c,S}^\top y_c$, $Q_S^c = y_c^\top y_c - n(z_S^c)^\top H_S^{-1} z_S^c$. Empty-model conventions: $\mathbb{R}^0$ is a point, $D_\emptyset = 1$, $Q_\emptyset = y^\top y$, $Q_\emptyset^c = y_c^\top y_c$, and all projections and smoothers of $\emptyset$ vanish. $a_0 = b_0 = 0.01$, $\nu = a_0 + n/2$; $\mathrm{IG}(v; a, b) = \frac{b^a}{\Gamma(a)} v^{-a-1} e^{-b/v}$ on $v > 0$. The prior weights are $w(k) = \frac{1}{2} / \binom{2p-1}{k}$, $W_K = \sum_{k \le K} \binom{p}{k} w(k)$, $\varpi(S) = w(|S|)/W_K$.

Contrast model at rate b. For $S \in \Omega$, $b > 0$, $\theta \in \mathbb{R}^S$, $v > 0$ put

$$g_{S,b}(\theta, v) = \mathcal{N}_{n-1}(y_c; X_{c,S}\theta, vI)\, \mathcal{N}_k\big(\theta; 0, nv(G_S + I)^{-1}\big)\, \mathrm{IG}(v; 0.51, b),$$

$m_b^c(S) = \iint g_{S,b}$, $I_b(E) = \sum_{S \in \Omega} \varpi(S) \iint_{E_S} g_{S,b}$, $Z_b^c = I_b(\mathrm{all}) = \sum_\Omega \varpi(S) m_b^c(S)$, $\Pi_b(E) = I_b(E)/Z_b^c$.

Statistical model. $y^r = \alpha^* \mathbf{1} + X^r \beta^* + \varepsilon$, $\varepsilon \sim \mathcal{N}_n(0, \sigma_*^2 I_n)$; $\mu = C X^r \beta^* = X\theta^*$, $S_0 = \operatorname{supp} \beta^*$, $s_0 = |S_0|$; $\varepsilon_c = U^\top \varepsilon$. Under G the input is valid $\mathbb{P}_0$-almost surely whenever X^r is a valid design: $Cy^r = \mu + C\varepsilon$ and $C\varepsilon$ has a non-degenerate Gaussian law on the $(n-1)$-dimensional space $\mathbf{1}^\perp$, so $\mathbb{P}_0(Cy^r = 0) = 0$. Hence $s_y > 0$ and $\Pi(\cdot \mid y)$ are defined almost surely. The constants $b_- = b_0 \sigma_*^2/2$, $b_+ = b_0(2(\sigma_*^2 + M) + 1)$, $\kappa = (b_+/b_-)^{0.51}$, and the event $\mathcal{B}_n = \{\sigma_*^2/2 \le s_y^2 \le 2(\sigma_*^2 + M) + 1\} = \{b_- \le b_0 s_y^2 \le b_+\}$ are as in `STATEMENT.md`.

2 Deterministic preliminaries

Throughout this section $n \ge 3$ is a fixed number, $a \ge 1$ and $k \ge 0$ are integers, $\mathsf{A} \in \mathbb{R}^{a \times k}$, $Y \in \mathbb{R}^a$, and

$$G = \mathsf{A}^\top \mathsf{A}, \quad H = (n+1)G + I_k, \quad D = \frac{\det H}{\det(G + I_k)},$$
$$Q = Y^\top Y - n(\mathsf{A}^\top Y)^\top H^{-1} \mathsf{A}^\top Y, \quad \mathsf{M} = n\mathsf{A}H^{-1}\mathsf{A}^\top.$$

P_A denotes the orthogonal projection onto the column span of A ($P_\mathsf{A} = 0$ if $k = 0$).

Lemma 2.1 (Gaussian slab integral)**.** *For every $v > 0$,*

$$\int_{\mathbb{R}^k} \mathcal{N}_a(Y; \mathsf{A}\theta, vI_a)\, \mathcal{N}_k\big(\theta; 0, nv(G+I)^{-1}\big)\, d\theta = (2\pi v)^{-a/2} D^{-1/2} e^{-Q/(2v)},$$

and the integrand, normalised in θ, is the density of $\mathcal{N}_k(nH^{-1}\mathsf{A}^\top Y,\ nvH^{-1})$. Moreover $Q = \min_\theta\{\|Y - \mathsf{A}\theta\|^2 + \theta^\top(G+I)\theta/n\} \ge 0$. (For $k = 0$ the integral is the value of the first factor, $D = 1$, $Q = Y^\top Y$.)

Proof. For $k = 0$ there is nothing to prove. Let $k \ge 1$; $G + I$ and H are positive definite. The exponent of the integrand is $-\frac{1}{2v} J(\theta)$ with

$$J(\theta) = \|Y - \mathsf{A}\theta\|^2 + \tfrac{1}{n}\theta^\top(G+I)\theta = Y^\top Y - 2\theta^\top \mathsf{A}^\top Y + \tfrac{1}{n}\theta^\top H \theta,$$

since $G + \frac{1}{n}(G+I) = \frac{1}{n}H$. With $m = nH^{-1}\mathsf{A}^\top Y$, completing the square gives $J(\theta) = (\theta - m)^\top \frac{H}{n}(\theta - m) + Y^\top Y - m^\top \frac{H}{n} m$, and $m^\top \frac{H}{n} m = n(\mathsf{A}^\top Y)^\top H^{-1} \mathsf{A}^\top Y$, so $J(\theta) = (\theta - m)^\top \frac{H}{n}(\theta - m) + Q$. This proves the minimum representation of Q and $Q \ge 0$, and shows that the normalised integrand is $\mathcal{N}_k(m, nvH^{-1})$. The constant in front of $e^{-J/(2v)}$ is $(2\pi v)^{-a/2}(2\pi)^{-k/2} \det(nv(G+I)^{-1})^{-1/2}$, and $\int e^{-(\theta-m)^\top H(\theta-m)/(2nv)} d\theta = (2\pi)^{k/2} \det(nvH^{-1})^{1/2}$. The product of the determinant factors is $\big((nv)^{-k} \det(G+I)\big)^{1/2} \big((nv)^k / \det H\big)^{1/2} = D^{-1/2}$. □

Lemma 2.2 (inverse-gamma integral)**.** *For $\mathfrak{a}, \mathfrak{b} > 0$: $\int_0^\infty v^{-\mathfrak{a}-1} e^{-\mathfrak{b}/v} dv = \Gamma(\mathfrak{a}) \mathfrak{b}^{-\mathfrak{a}}$.*

Proof. Substitute $t = \mathfrak{b}/v$: the integral becomes $\mathfrak{b}^{-\mathfrak{a}} \int_0^\infty t^{\mathfrak{a}-1} e^{-t} dt$. □

Lemma 2.3 (smoother and determinant)**.** *Let $h(\lambda) = \frac{n\lambda}{(n+1)\lambda+1}$ and $f(\lambda) = \frac{(n+1)\lambda+1}{\lambda+1}$ for $\lambda \geq 0$.*

(a) $\mathsf{M} = h(\mathsf{A}\mathsf{A}^\top)$ (functional calculus), and $Q = Y^\top (I - \mathsf{M}) Y$.

(b) $0 \preceq \mathsf{M} \preceq \frac{n}{n+1} P_{\mathsf{A}}$; hence $I - \mathsf{M} \succeq I - P_{\mathsf{A}}$ and $I - \mathsf{M} \succeq \frac{1}{n+1} I$, so $Q \geq \|(I - P_{\mathsf{A}})Y\|^2$ and $Q \geq \|Y\|^2/(n+1)$.

(c) If $k \geq 1$ and $\lambda_{\min}(G) \geq g > 0$, then $\mathsf{M} \succeq \frac{n}{n+1}\big(1 - \frac{1}{(n+1)g}\big) P_{\mathsf{A}}$ and $I - \mathsf{M} \preceq (I - P_{\mathsf{A}}) + \gamma P_{\mathsf{A}}$ with $\gamma = \frac{1}{n+1}\big(1 + \frac{1}{g}\big)$.

(d) If A' is obtained from A by adjoining columns, then $\mathsf{M}_{\mathsf{A}} \preceq \mathsf{M}_{\mathsf{A}'}$; hence $Q_{\mathsf{A}} \geq Q_{\mathsf{A}'}$.

(e) $D = \prod_{i=1}^k f(\lambda_i(G))$; in particular $1 \leq D \leq (n+1)^k$. If A' is A with $j \geq 1$ adjoined columns, $G' = \mathsf{A}'^\top \mathsf{A}'$ and $\lambda_{\min}(G') \geq g > 0$, then $D'/D \geq f(g)^j \geq \big[(n+1)(1+1/g)^{-1}\big]^j$.

Proof. (a) For $c = n+1$, $\mathsf{A}(c\mathsf{A}^\top\mathsf{A} + I) = (c\mathsf{A}\mathsf{A}^\top + I)\mathsf{A}$, so $\mathsf{A}(c\mathsf{A}^\top\mathsf{A} + I)^{-1} = (c\mathsf{A}\mathsf{A}^\top + I)^{-1}\mathsf{A}$ and $\mathsf{M} = n(c\mathsf{A}\mathsf{A}^\top + I)^{-1}\mathsf{A}\mathsf{A}^\top = h(\mathsf{A}\mathsf{A}^\top)$, the two factors commuting. Then $Q = Y^\top Y - Y^\top \mathsf{M} Y$ by definition of M.

(b) h is increasing, $h(0) = 0$ and $h < \frac{n}{n+1}$. The eigenvectors of $\mathsf{A}\mathsf{A}^\top$ with nonzero eigenvalue span the column space of A, and M vanishes on its orthogonal complement; so the eigenvalues of M lie in $[0, \frac{n}{n+1})$ on the column space and are 0 off it. Thus $\mathsf{M} \preceq \frac{n}{n+1} P_{\mathsf{A}} \preceq P_{\mathsf{A}}$, and $I - \mathsf{M} \succeq I - \frac{n}{n+1} P_{\mathsf{A}} \succeq \frac{1}{n+1} I$.

(c) The nonzero eigenvalues of $\mathsf{A}\mathsf{A}^\top$ are those of G, all $\geq g$, and their eigenvectors span the column space. Since $h(\lambda) = \frac{n}{n+1}\big(1 - \frac{1}{(n+1)\lambda+1}\big) \geq \frac{n}{n+1}\big(1 - \frac{1}{(n+1)g}\big)$ for $\lambda \geq g$, the first claim follows. For the second: on the column space, $1 - h(\lambda) = \frac{\lambda+1}{(n+1)\lambda+1} \leq \frac{\lambda+1}{(n+1)\lambda} = \frac{1}{n+1}\big(1 + \frac{1}{\lambda}\big) \leq \gamma$; off it, $I - \mathsf{M}$ acts as the identity, which is $I - P_{\mathsf{A}}$ there.

(d) $h(\lambda) = \frac{n}{n+1}\big(1 - ((n+1)\lambda+1)^{-1}\big)$. If $0 \preceq B_1 \preceq B_2$ then $((n+1)B_2 + I)^{-1} \preceq ((n+1)B_1 + I)^{-1}$, so $h(B_1) \preceq h(B_2)$. Apply this with $B_1 = \mathsf{A}\mathsf{A}^\top \preceq \mathsf{A}'\mathsf{A}'^\top = B_2$; then use (a).

(e) H and $G + I$ are polynomials in G, so $D = \prod_i f(\lambda_i(G))$; f is increasing with $f(0) = 1$ and $f < n + 1$. Order eigenvalues increasingly. G is a principal submatrix of G' (columns of A first), so Cauchy interlacing gives $\lambda_i(G) \leq \lambda_{i+j}(G')$ for $1 \leq i \leq k$, whence $D \leq \prod_{i=1}^k f(\lambda_{i+j}(G'))$ and $D'/D \geq \prod_{i=1}^j f(\lambda_i(G')) \geq f(g)^j$. Finally $f(g) \geq \frac{(n+1)g}{g+1} = (n+1)(1+1/g)^{-1}$. □

Lemma 2.4 (the capped BB$(1,p)$ prior)**.** *(a) $w(k) = B(1+k, 2p-k)/B(1,p)$ for $0 \leq k \leq p$, and $\sum_{k=0}^p \binom{p}{k} w(k) = 1$; hence $\frac{1}{2} = w(0) \leq W_K \leq 1$.*

(b) $w(k+1)/w(k) = (k+1)/(2p-k-1)$ for $0 \leq k \leq p-1$.

(c) For every integer m: $\varpi(\{S \in \Omega : |S| > m\}) \leq 2^{1-m}$.

(d) If $s_0 \leq K$, then $\varpi(S_0) \geq \frac{1}{2}(2p)^{-s_0}$.

(e) For $j \geq 1$ with $s_0 + j \leq p$: $\binom{p-s_0}{j} \frac{w(s_0+j)}{w(s_0)} \leq 2\binom{s_0+j}{j} 2^{-j} \leq 2(j+1)^{s_0} 2^{-j}$.

Proof. (a) $B(1+k, 2p-k) = k!(2p-k-1)!/(2p)!$ and $B(1,p) = 1/p$, so the ratio is $\frac{k!(2p-1-k)!}{2(2p-1)!} = \frac{1}{2}\binom{2p-1}{k}^{-1} = w(k)$. By the binomial theorem, $\sum_k \binom{p}{k} \int_0^1 t^k (1-t)^{p-k} \frac{(1-t)^{p-1}}{B(1,p)} dt = \int_0^1 \frac{(1-t)^{p-1}}{B(1,p)} dt = 1$, and the k-th integral is $B(1+k, 2p-k)/B(1,p)$. The bounds on W_K follow since all terms are positive and the $k = 0$ term is $w(0) = \frac{1}{2}$.

(b) $w(k+1)/w(k) = \binom{2p-1}{k}/\binom{2p-1}{k+1} = (k+1)/(2p-1-k)$.

(c) Put $q_k = \binom{p}{k} w(k)$. By (b), $q_{k+1}/q_k = \frac{p-k}{k+1} \cdot \frac{k+1}{2p-k-1} = \frac{p-k}{2p-k-1}$, which is $\le \frac{1}{2}$ iff $k \ge 1$. Also $q_1 = \frac{p}{2(2p-1)} \le \frac{1}{2}$. Hence $q_k \le 2^{-(k-1)} q_1 \le 2^{-k}$ for $1 \le k \le p$, and for $m \ge 0$, $\sum_{k>m} q_k \le \sum_{k \ge m+1} 2^{-k} = 2^{-m}$. Then $\varpi(|S| > m) = \sum_{m<k\le K} q_k / W_K \le 2 \cdot 2^{-m}$ by (a). For $m < 0$ the bound exceeds 1.

(d) $\varpi(S_0) = w(s_0)/W_K \ge w(s_0) = \frac{1}{2}\binom{2p-1}{s_0}^{-1} \ge \frac{1}{2}(2p-1)^{-s_0} \ge \frac{1}{2}(2p)^{-s_0}$.

(e) Write both factors as products over $i = 0, \dots, j-1$, using (b):

$$\binom{p-s_0}{j} \frac{w(s_0+j)}{w(s_0)} = \prod_{i=0}^{j-1} \frac{p-s_0-i}{i+1} \cdot \frac{s_0+i+1}{2p-s_0-i-1} = \binom{s_0+j}{j} \prod_{i=0}^{j-1} \frac{p-s_0-i}{2p-s_0-i-1}.$$

The denominators are $\ge 2p - s_0 - j \ge p \ge 1$. A factor is $\le \frac{1}{2}$ iff $s_0 + i \ge 1$; this fails only for $s_0 = 0, i = 0$, where the factor is $p/(2p-1) \le 1$. So the product is $\le 2 \cdot 2^{-j}$. Finally $\binom{s_0+j}{j} = \prod_{i=1}^{s_0} \frac{j+i}{i} \le (j+1)^{s_0}$. □

Lemma 2.5 (chi-square facts)**.** *Let χ^2_r denote a chi-square variable with $r \ge 0$ degrees of freedom ($\chi^2_0 \equiv 0$).*

(a) If $r \le d$ and $x \ge 0$, then $\mathbb{P}(\chi^2_r \ge 2d \log 2 + 4x) \le e^{-x}$.

(b) For $d \ge 1$ and $t > 0$: $\mathbb{P}(|\chi^2_d - d| > t) \le 2d/t^2$, and $\mathbb{E}|\chi^2_d - d| \le \sqrt{2d}$.

(c) For $d \ge 1$ and $\tau > 0$: $\mathbb{E}\big[e^{\chi^2_d/2} \mathbf{1}\{\chi^2_d \le \tau\}\big] = \frac{(\tau/2)^{d/2}}{\Gamma(d/2+1)} \le \big(\frac{e\tau}{d}\big)^{d/2}$.

Proof. (a) $\mathbb{E}e^{\chi^2_r/4} = (1-\frac{1}{2})^{-r/2} = 2^{r/2} \le 2^{d/2}$, so by Markov's inequality the probability is at most $2^{d/2} e^{-(2d\log 2 + 4x)/4} = e^{-x}$. (b) $\mathrm{Var}(\chi^2_d) = 2d$; Chebyshev and Cauchy–Schwarz. (c) The density of χ^2_d is $x^{d/2-1}e^{-x/2}/(2^{d/2}\Gamma(d/2))$, so the expectation is $\int_0^\tau x^{d/2-1}dx/(2^{d/2}\Gamma(d/2)) = \tau^{d/2}/(2^{d/2}\frac{d}{2}\Gamma(\frac{d}{2}))$. The bound uses $\Gamma(x+1) \ge \int_x^\infty t^x e^{-t} dt \ge x^x e^{-x}$. □

3 T01(i): the contrast identity

Proposition 3.1 (closed forms)**.** *For every valid input, $S \in \Omega$ and $b > 0$:*

$$\begin{aligned} m(S) &:= \iint \mathcal{N}_n(y; X_S\beta, \sigma^2 I_n)\, \mathcal{N}_k\big(\beta; 0, n\sigma^2 (G_S + I)^{-1}\big) \mathrm{IG}(\sigma^2; a_0, b_0)\, d\beta\, d\sigma^2 \\ &= \frac{(2b_0)^{a_0}\Gamma(\nu)}{\pi^{n/2}\Gamma(a_0)} D_S^{-1/2} (2b_0 + Q_S)^{-\nu}, \qquad (2) \\ m^c_b(S) &= (2\pi)^{-(n-1)/2} \frac{b^{0.51}\Gamma(\nu)}{\Gamma(0.51)} D_S^{-1/2} \big(b + Q^c_S/2\big)^{-\nu}. \qquad (3) \end{aligned}$$

So the integral in (2) is the method's $m(S)$. Under the contrast model at rate b given S and y_c, $v \sim \mathrm{IG}(\nu, b + Q^c_S/2)$ and $\theta_S \mid v \sim \mathcal{N}_k(nH_S^{-1} z^c_S, nvH_S^{-1})$.

Proof. By Tonelli, integrate β first using Lemma 2.1 with $a = n$, $\mathsf{A} = X_S$, $Y = y$. This gives $(2\pi\sigma^2)^{-n/2} D_S^{-1/2} e^{-Q_S/(2\sigma^2)}$. Then by Lemma 2.2 with $\mathfrak{a} = a_0 + n/2 = \nu$ and $\mathfrak{b} = b_0 + Q_S/2 > 0$,

$$m(S) = (2\pi)^{-n/2} D_S^{-1/2} \frac{b_0^{a_0}}{\Gamma(a_0)} \Gamma(\nu)(b_0 + Q_S/2)^{-\nu} = \frac{(2b_0)^{a_0}\Gamma(\nu)}{\pi^{n/2}\Gamma(a_0)} D_S^{-1/2} (2b_0 + Q_S)^{-\nu},$$

using $(b_0+Q/2)^{-\nu}=2^{\nu}(2b_0+Q)^{-\nu}$ and $2^{-n/2+\nu}=2^{a_0}$. For (3), use Lemma 2.1 with $a=n-1$, $\mathsf{A}=X_{c,S}$ (so $\mathsf{A}^\top\mathsf{A}=G_S$), $Y=y_c$; then Lemma 2.2 with $\mathfrak{a}=0.51+(n-1)/2=\nu$ and $\mathfrak{b}=b+Q_S^c/2$. The conditional law of θ_S is the normalised integrand of Lemma 2.1. The v-marginal is proportional to $v^{-(n-1)/2}e^{-Q_S^c/(2v)}v^{-1.51}e^{-b/v}=v^{-\nu-1}e^{-(b+Q_S^c/2)/v}$. □

Theorem 3.2 (T01(i))**.** *For every valid input, the joint transformed posterior of (S,θ_S,v) equals Π_b with $b=b_0s_y^2$. For every $S\in\Omega$, $m(S)=c_n(s_y)\,m^c_{b_0s_y^2}(S)$ with $c_n(s)=s^{n-1}\Gamma(0.51)/(\sqrt{2\pi b_0}\,\Gamma(0.01))$. Moreover $z_S^c=s_yz_S$, $Q_S^c=s_y^2Q_S$, $Z=c_n(s_y)Z^c_{b_0s_y^2}$,*

$$\mathrm{BF}(S)=\frac{m^c_{b_0s_y^2}(S)}{m^c_{b_0s_y^2}(\emptyset)},\qquad \pi(S)=\frac{\varpi(S)m^c_{b_0s_y^2}(S)}{Z^c_{b_0s_y^2}},$$

and the conditionals are $v\mid S,y\sim\mathrm{IG}(\nu,b_0s_y^2+Q_S^c/2)$ and $\theta_S\mid v,S,y\sim\mathcal{N}_k(nH_S^{-1}z_S^c,nvH_S^{-1})$.

Proof. Write $s=s_y$ and $b=b_0s^2$. By (1), $z_S^c=X_S^\top UU^\top y^r=X_S^\top Cy^r=s\,X_S^\top y=s\,z_S$ and $y_c^\top y_c=s^2y^\top y$, hence $Q_S^c=s^2Q_S$.

The v3 joint posterior. The working model of `STATEMENT.md` §1 has, at the observed y, the unnormalised joint posterior density

$$p(S,\beta,\sigma^2)=\varpi(S)\,\mathcal{N}_n(y;X_S\beta,\sigma^2I_n)\,\mathcal{N}_k\big(\beta;0,n\sigma^2(G_S+I)^{-1}\big)\mathrm{IG}(\sigma^2;a_0,b_0),$$

with respect to counting measure on Ω times Lebesgue measure on $\mathbb{R}^S\times(0,\infty)$. Its total mass is $\sum_\Omega\varpi(S)m(S)=Z\in(0,\infty)$ by Proposition 3.1. Its S-marginal is π.

Change of variables. The map $(\beta,\sigma^2)\mapsto(\theta,v)=(s\beta,s^2\sigma^2)$ is a linear bijection of $\mathbb{R}^S\times(0,\infty)$ with inverse Jacobian determinant s^{-k-2}. So the image law has unnormalised density $q(S,\theta,v)=p(S,\theta/s,v/s^2)\,s^{-k-2}$. We evaluate the three factors. By (1), $y-X_S\theta/s=U(y_c-X_{c,S}\theta)/s$ and U is an isometry, so

$$\mathcal{N}_n(y;X_S\theta/s,(v/s^2)I_n)=s^n(2\pi v)^{-n/2}\exp\Big\{-\frac{\|y_c-X_{c,S}\theta\|^2}{2v}\Big\}.$$

Next, $\mathcal{N}_k(\theta/s;0,n(v/s^2)(G_S+I)^{-1})=s^k\mathcal{N}_k(\theta;0,nv(G_S+I)^{-1})$. Finally,

$$\mathrm{IG}(v/s^2;a_0,b_0)=\frac{b_0^{a_0}}{\Gamma(a_0)}\Big(\frac{v}{s^2}\Big)^{-a_0-1}e^{-b_0s^2/v}=s^2\,\mathrm{IG}(v;a_0,b_0s^2).$$

Hence $q(S,\theta,v)=\varpi(S)s^n(2\pi v)^{-n/2}e^{-\|y_c-X_{c,S}\theta\|^2/(2v)}\mathcal{N}_k(\theta;0,nv(G_S+I)^{-1})\mathrm{IG}(v;a_0,b)$. Now

$$(2\pi v)^{-n/2}\mathrm{IG}(v;a_0,b)=(2\pi v)^{-(n-1)/2}(2\pi)^{-1/2}\frac{b^{a_0}}{\Gamma(a_0)}v^{-a_0-3/2}e^{-b/v}$$

$$=(2\pi v)^{-(n-1)/2}\frac{\Gamma(0.51)\,b^{a_0-0.51}}{\sqrt{2\pi}\,\Gamma(a_0)}\mathrm{IG}(v;0.51,b),$$

and $b^{a_0-0.51}=(b_0s^2)^{-1/2}=b_0^{-1/2}s^{-1}$. Therefore

$$q(S,\theta,v)=c_n(s)\,\varpi(S)\,g_{S,b}(\theta,v),\qquad c_n(s)=\frac{s^{n-1}\Gamma(0.51)}{\sqrt{2\pi b_0}\,\Gamma(0.01)}.\tag{4}$$

Conclusions. The change of variables preserves integrals. Integrating (4) over (θ,v) gives $\varpi(S)m(S)=c_n(s)\varpi(S)m_b^c(S)$, so $m(S)=c_n(s)m_b^c(S)$ for all $S\in\Omega$. Summing gives $Z=c_n(s)Z_b^c$. The normalised image law is $q/Z=\varpi(S)g_{S,b}/Z_b^c$, the density of Π_b. The expressions for BF and π follow because $c_n(s)$ does not depend on S. The conditionals are those of the contrast model at rate b (Proposition 3.1). □

4 T01(ii) and (iv): envelope and rate derivative

Theorem 4.1 (T01(ii)). *Let $0 < b_1 \le b \le b_2$ and $\kappa_{12} = (b_2/b_1)^{0.51}$. For every $y_c \in \mathbb{R}^{n-1}$, $S \in \Omega$ and measurable parameter event E:*

$$\kappa_{12}^{-1} m_{b_2}^c(S) \le m_b^c(S) \le \kappa_{12}\, m_{b_1}^c(S), \qquad \kappa_{12}^{-1} I_{b_2}(E) \le I_b(E) \le \kappa_{12} I_{b_1}(E), \qquad \Pi_b(E) \le \kappa_{12}^2 \frac{I_{b_1}(E)}{Z_{b_2}^c}.$$

With $b_1 = b_-$, $b_2 = b_+$ one has $\kappa_{12} = \kappa$.

Proof. For $v > 0$ and $b, b' > 0$, $\mathrm{IG}(v; 0.51, b)/\mathrm{IG}(v; 0.51, b') = (b/b')^{0.51} e^{-(b-b')/v}$. With $b' = b_1 \le b$, the ratio is at most $(b/b_1)^{0.51} \le \kappa_{12}$. With $b' = b_2 \ge b$, it is at least $(b/b_2)^{0.51} \ge (b_1/b_2)^{0.51} = \kappa_{12}^{-1}$. Hence $\kappa_{12}^{-1} g_{S,b_2} \le g_{S,b} \le \kappa_{12} g_{S,b_1}$ pointwise, since the remaining factors of $g_{S,\cdot}$ are nonnegative and do not depend on the rate. Integrating over all of $\mathbb{R}^S \times (0,\infty)$, or over E_S, and summing with weights $\varpi(S)$ gives the first two chains. Finally $Z_b^c = I_b(\mathrm{all}) \ge \kappa_{12}^{-1} Z_{b_2}^c > 0$, so $\Pi_b(E) = I_b(E)/Z_b^c \le \kappa_{12} I_{b_1}(E)/(\kappa_{12}^{-1} Z_{b_2}^c)$. The argument is pointwise in b and y_c, so b may be evaluated at any data-dependent value in $[b_1, b_2]$. □

Theorem 4.2 (T01(iv)). *For fixed contrast data, $S, S' \in \Omega$ and $b > 0$: $\partial_b \log[m_b^c(S)/m_b^c(S')] = 2\nu(Q_S^c - Q_{S'}^c)/((2b + Q_S^c)(2b + Q_{S'}^c))$.*

Proof. By (3), $\log[m_b^c(S)/m_b^c(S')] = -\frac{1}{2}\log(D_S/D_{S'}) - \nu \log(b + Q_S^c/2) + \nu \log(b + Q_{S'}^c/2)$. Both arguments of the logarithms are positive because $Q^c \ge 0$. Differentiating gives $\nu\big[\frac{1}{b+Q_{S'}^c/2} - \frac{1}{b+Q_S^c/2}\big] = \nu \frac{(Q_S^c - Q_{S'}^c)/2}{(b+Q_S^c/2)(b+Q_{S'}^c/2)}$, which equals the claim after multiplying the numerator and denominator by 4. □

5 T01(iii): the scale event

Theorem 5.1 (T01(iii)). *Under G, $\varepsilon_c = U^\top \varepsilon \sim \mathcal{N}_{n-1}(0, \sigma_*^2 I_{n-1})$ exactly and $y_c = X_c\theta^* + \varepsilon_c$. If also $\|\mu\|^2 \le nM$ (part of R), then with $\Delta_n := s_y^2 - (\sigma_*^2 + \|\mu\|^2/(n-1))$,*

$$\mathbb{E}_0 \Delta_n = 0, \qquad \mathbb{E}_0 \Delta_n^2 \le \frac{2\sigma_*^4 + 6\sigma_*^2 M}{n-1}, \qquad \mathbb{P}_0(\mathcal{B}_n^c) \le \frac{8 + 24M/\sigma_*^2}{n-1}.$$

In particular $\Delta_n = O_{\mathbb{P}_0}(n^{-1/2})$ and $\mathbb{P}_0(\mathcal{B}_n) \to 1$.

Proof. $U^\top \varepsilon$ is Gaussian with mean 0 and covariance $\sigma_*^2 U^\top U = \sigma_*^2 I_{n-1}$. Since $U^\top \mathbf{1} = 0$ and $U^\top = U^\top C$, $y_c = U^\top X^r \beta^* + U^\top \varepsilon = U^\top \mu + \varepsilon_c$, and $U^\top \mu = U^\top X \theta^* = X_c \theta^*$.

Next, $Cy^r = \mu + C\varepsilon$, so $(n-1)s_y^2 = \|\mu\|^2 + 2\mu^\top C\varepsilon + \|C\varepsilon\|^2$. Here $\mu^\top C\varepsilon = \mu^\top \varepsilon$ because $C\mu = \mu$. Also $\|C\varepsilon\|^2 = \varepsilon^\top U U^\top \varepsilon = \|\varepsilon_c\|^2 = \sigma_*^2 E'$ with $E' \sim \chi_{n-1}^2$. Hence

$$\Delta_n = \sigma_*^2 \Big(\frac{E'}{n-1} - 1 \Big) + \frac{2\mu^\top \varepsilon}{n-1}.$$

Both terms have mean 0. $\mathrm{Var}(E') = 2(n-1)$ and $\mathrm{Var}(\mu^\top \varepsilon) = \sigma_*^2 \|\mu\|^2$. The cross moment $\mathbb{E}[(E' - (n-1))\mu^\top \varepsilon]$ vanishes, because $\varepsilon \mapsto -\varepsilon$ preserves the law of ε, leaves E' unchanged and flips the sign of $\mu^\top \varepsilon$. Thus $\mathbb{E}\Delta_n^2 = \frac{2\sigma_*^4}{n-1} + \frac{4\sigma_*^2 \|\mu\|^2}{(n-1)^2} \le \frac{2\sigma_*^4}{n-1} + \frac{4\sigma_*^2 Mn}{(n-1)^2}$, and $n/(n-1) \le \frac{3}{2}$ for $n \ge 3$. By Chebyshev, $\mathbb{P}_0(\sqrt{n}|\Delta_n| > L) \le \frac{3}{2}(2\sigma_*^4 + 6\sigma_*^2 M)/L^2$ for every $L > 0$, which is the $O_{\mathbb{P}_0}(n^{-1/2})$ claim.

For $\mathcal{B}_n$, note $\sigma_*^2 \le \sigma_*^2 + \|\mu\|^2/(n-1) \le \sigma_*^2 + \frac{3}{2}M$. If $s_y^2 < \sigma_*^2/2$ then $\Delta_n < -\sigma_*^2/2$. If $s_y^2 > 2(\sigma_*^2 + M) + 1$ then $\Delta_n > \sigma_*^2 + \frac{M}{2} + 1 > \sigma_*^2/2$. So $\mathcal{B}_n^c \subseteq \{|\Delta_n| > \sigma_*^2/2\}$, and Chebyshev gives $\mathbb{P}_0(\mathcal{B}_n^c) \le 4\mathbb{E}\Delta_n^2/\sigma_*^4$, which is the stated bound. □

6 The transfer rule

Theorem 6.1 (T01 (T)(a): mixed-endpoint transfer)**.** *For every valid input on the event $\mathcal{B}_n$, and every measurable parameter event E,*

$$\Pi(E \mid y) \le \kappa^2 \frac{I_{b_-}(E)}{Z^c_{b_+}} \le \kappa^2 \frac{I_{b_-}(E)}{\varpi(S_0) m^c_{b_+}(S_0)},$$

the last inequality when $S_0 \in \Omega$. Under G and R, let E_n be events (deterministic, or data-dependent with $\Pi(E_n \mid y)$ and $I_{b_-}(E_n)/Z^c_{b_+}$ measurable) and $t_n > 0$ deterministic. If $\mathbb{P}_0(I_{b_-}(E_n)/Z^c_{b_+} \le t_n) \to 1$, then $\mathbb{P}_0(\Pi(E_n \mid y) \le \kappa^2 t_n) \to 1$. If $I_{b_-}(E_n)/Z^c_{b_+} = O_{\mathbb{P}_0}(t_n)$, then $\Pi(E_n \mid y) = O_{\mathbb{P}_0}(t_n)$.

Proof. By Theorem 3.2, $\Pi(E \mid y) = \Pi_b(E)$ with $b = b_0 s_y^2$. On $\mathcal{B}_n$ this number lies in $[b_-, b_+]$. Theorem 4.1 with $b_1 = b_-$, $b_2 = b_+$ is a deterministic inequality, valid for every y_c and every $b \in [b_-, b_+]$, so it applies at the observed pair $(y_c, b_0 s_y^2)$ and gives the first inequality. The second holds because $Z^c_{b_+} \ge \varpi(S_0) m^c_{b_+}(S_0)$. For the consequences, the first inequality holds pointwise on $\mathcal{B}_n$, so $\mathbb{P}_0(\Pi(E_n \mid y) > \kappa^2 x) \le \mathbb{P}_0(\mathcal{B}_n^c) + \mathbb{P}_0(I_{b_-}(E_n)/Z^c_{b_+} > x)$ for every $x > 0$, and $\mathbb{P}_0(\mathcal{B}_n^c) \to 0$ by Theorem 5.1. With $x = t_n$ this gives the first consequence. With $x = Lt_n$, $\limsup_n \mathbb{P}_0(\Pi(E_n \mid y) > \kappa^2 L t_n) \le \limsup_n \mathbb{P}_0(I_{b_-}(E_n)/Z^c_{b_+} > L t_n)$, which tends to 0 as $L \to \infty$; this is the second. □

For a model $S \in \Omega$ we write $\Pi_r(S)$ for the posterior probability of the model event $\{S\} \times \mathbb{R}^S \times (0, \infty)$. Statements (b1) and (b2) are worst cases over inputs; they are not statements about $\mathbb{P}_0$.

Theorem 6.2 (T01 (T)(b1): two fixed rates)**.** *Let $0 < r_1 < r_2$ be fixed. There are valid inputs with $p = 1$ ($n \ge 3$) such that $\Pi_{r_1}(\{1\}) \to 1$ and $\Pi_{r_2}(\{1\}) \to 0$, hence $\Pi_{r_1}(\emptyset) \to 0$ and $\Pi_{r_2}(\emptyset) \to 1$. Consequently, for distinct fixed rates $b \ne b'$ there is no function g with $g(x) \to 0$ as $x \downarrow 0$, independent of n and the data, such that $\Pi_{b'}(E) \le g(\Pi_b(E))$ for all valid inputs and all events E. (These inputs have $s_y^2 \to 0$, so they lie outside $\mathcal{B}_n$ for large n; Theorem 6.3 is the version on $\mathcal{B}_n$.)*

Proof. Fix a unit vector $e \in \mathbb{R}^{n-1}$ and put $X^r = Ue$. It is centred and nonzero, with sample variance $1/(n-1)$, so $X_c = \sqrt{n-1}\, e$. Here $K = 1$, $\Omega = \{\emptyset, \{1\}\}$, and $w(0) = w(1) = \frac{1}{2}$ by Lemma 2.4(a). Fix $\gamma \in (r_1, r_2)$, let $q_n = 2\gamma \log n / n$, and put $y^r = U\sqrt{q_n}\, e$, which is centred and nonzero; $y_c = \sqrt{q_n}\, e$. Then $G_{\{1\}} = n - 1$, $H_{\{1\}} = n^2$, $D_{\{1\}} = n$, $z^c_{\{1\}} = \sqrt{(n-1)q_n}$, $Q^c_\emptyset = q_n$ and $Q^c_{\{1\}} = q_n - n(n-1)q_n/n^2 = q_n/n$. By (3), for a fixed rate $r > 0$,

$$O_r := \frac{\Pi_r(\{1\})}{\Pi_r(\emptyset)} = n^{-1/2}\Big(\frac{2r + q_n}{2r + q_n/n}\Big)^\nu = n^{-1/2}(1 + u_r)^\nu, \qquad u_r = \frac{q_n(1 - 1/n)}{2r + q_n/n}.$$

Here $u_r \ge 0$, $u_r = O(\log n/n)$, and $\nu u_r = \gamma \log n \cdot \ell_n'(r)$ with $\ell_n'(r) = \frac{(1+0.02/n)(1-1/n)}{2r+q_n/n} \to \frac{1}{2r}$. Using $u - u^2/2 \le \log(1+u) \le u$, $\log O_r = \log n\,[\gamma \ell_n'(r) - \frac{1}{2}] + \rho$ with $|\rho| \le \nu u_r^2/2 = O(\log^2 n/n)$. Since $\gamma > r_1$, $O_{r_1} \to \infty$; since $\gamma < r_2$, $O_{r_2} \to 0$. As $\Pi_r(\{1\}) = O_r/(1 + O_r) = 1 - \Pi_r(\emptyset)$, the limits follow. For the consequence, suppose such a g existed. If $b' < b$, apply the construction with $r_1 = b'$, $r_2 = b$ and $E = \{1\}$: then $\Pi_b(E) \to 0$, so $g(\Pi_b(E)) \to 0$, while $\Pi_{b'}(E) \to 1$. If $b' > b$, use $r_1 = b$, $r_2 = b'$ and $E = \emptyset$: then $\Pi_b(E) \to 0$ while $\Pi_{b'}(E) \to 1$. Either case is a contradiction. □

Theorem 6.3 (T01 (T)(b2): on the scale event)**.** *Fix $\sigma_*^2 > 0$, $M \ge 0$ (hence b_-, b_+) and $s^2 \in (\sigma_*^2/2,\, 2(\sigma_*^2 + M) + 1)$, and put $b_* = b_0 s^2 \in (b_-, b_+)$. There are valid inputs with $p = 2$ and $s_y^2 = s^2$ for every $n \ge 4$ (so $\mathcal{B}_n$ holds for every n) such that $\Pi_{b_-}(\{1\}) \to 0$ and $\Pi(\{1\} \mid y) \to 1$ as $n \to \infty$. There are also valid inputs with the same properties such that $\Pi_{b_+}(\{1,2\}) \to 0$ and $\Pi(\{1,2\} \mid y) \to 1$. Consequently, for $b \in \{b_-, b_+\}$ there is no constant C, independent of n and of the data, such that $\Pi(E \mid y) \le C\,\Pi_b(E)$ for all valid inputs in $\mathcal{B}_n$ and all events E.*

Proof. Construction. Let $n \geq 4$, so $K = 2$ and Ω consists of all four subsets of $\{1,2\}$. Fix orthonormal $e_1, e_2 \in \mathbb{R}^{n-1}$ and put $X^r = U[e_1, e_2]$. Its columns are centred ($\mathbf{1}^\top U = 0$), nonzero, hence non-constant, with sample variance $1/(n-1)$. So $X = CX^r D_x^{-1} = \sqrt{n-1}\,U[e_1, e_2]$ and $X_c = \sqrt{n-1}[e_1, e_2]$. Given $\gamma > 0$ (a constant chosen below), let $\lambda_n = 2\gamma \log n/n$ and $\alpha_n = (n-1)s^2 - \lambda_n$. If $\alpha_n \geq 0$ put $y^r = U(\sqrt{\alpha_n}e_1 + \sqrt{\lambda_n}e_2)$; otherwise (finitely many n) put $y^r = U\sqrt{(n-1)s^2}\,e_1$. The latter n play no role in the limits. Then y^r is centred and nonzero, so it is non-constant, $y_c = U^\top y^r$, and $s_y^2 = \|y_c\|^2/(n-1) = s^2$ by (1).

Model quantities (for $\alpha_n \geq 0$; write $\alpha = \alpha_n$, $\lambda = \lambda_n$). $G_{\{j\}} = n-1$ and $G_{\{1,2\}} = (n-1)I_2$. So $H = n^2$, resp. $n^2 I_2$, and $D_{\{1\}} = D_{\{2\}} = n^2/n = n$, $D_{\{1,2\}} = n^2$. Also $z^c_{\{1\}} = \sqrt{(n-1)\alpha}$, $z^c_{\{2\}} = \sqrt{(n-1)\lambda}$ and $z^c_{\{1,2\}} = \sqrt{n-1}(\sqrt{\alpha}, \sqrt{\lambda})$. Hence

$$Q^c_\emptyset = \alpha + \lambda = (n-1)s^2, \quad Q^c_{\{1\}} = \tfrac{\alpha}{n} + \lambda, \quad Q^c_{\{2\}} = \alpha + \tfrac{\lambda}{n}, \quad Q^c_{\{1,2\}} = \tfrac{\alpha+\lambda}{n} = \tfrac{(n-1)s^2}{n}.$$

By Lemma 2.4(a) with $p = 2$: $w(0) = \frac{1}{2}$ and $w(1) = w(2) = \frac{1}{6}$. By (3), for any fixed rate $r > 0$, $\Pi_r(S) \propto w(|S|) D_S^{-1/2} (2r + Q^c_S)^{-\nu}$. Define the odds against $\{1\}$:

$$O_r(12) := \frac{\Pi_r(\{1,2\})}{\Pi_r(\{1\})} = n^{-1/2}(1+u_r)^\nu, \qquad u_r := \frac{Q^c_{\{1\}} - Q^c_{\{1,2\}}}{2r + Q^c_{\{1,2\}}} = \frac{\lambda(1-1/n)}{2r + s^2(1-1/n)},$$
$$O_r(\emptyset) := \frac{\Pi_r(\emptyset)}{\Pi_r(\{1\})} = 3n^{1/2}\Big(\frac{2r + \alpha/n + \lambda}{2r + (n-1)s^2}\Big)^\nu, \qquad O_r(2) := \frac{\Pi_r(\{2\})}{\Pi_r(\{1\})} = \Big(\frac{2r + \alpha/n + \lambda}{2r + \alpha + \lambda/n}\Big)^\nu.$$

Limits. Since $\alpha/n \leq s^2$, $\lambda \to 0$ and $\alpha \to \infty$, both bases in $O_r(\emptyset)$ and $O_r(2)$ are at most $\frac{1}{2}$ for large n. So $O_r(\emptyset) \leq 3n^{1/2}2^{-\nu} \to 0$ and $O_r(2) \leq 2^{-\nu} \to 0$. For $O_r(12)$: $u_r \geq 0$, $u_r = O(\log n/n)$ and

$$\nu u_r = \gamma \log n \cdot \ell_n(r), \qquad \ell_n(r) = \frac{(1 + 0.02/n)(1-1/n)}{2r + s^2(1-1/n)} \to \frac{1}{2r+s^2}.$$

Using $u - u^2/2 \leq \log(1+u) \leq u$ for $u \geq 0$, $\log O_r(12) = \log n\,\big[\gamma \ell_n(r) - \frac{1}{2}\big] + \rho$ with $|\rho| \leq \nu u_r^2/2 = O(\log^2 n/n) \to 0$. Hence $O_r(12) \to \infty$ if $\gamma > (2r+s^2)/2$ and $O_r(12) \to 0$ if $\gamma < (2r+s^2)/2$. Since

$$\Pi_r(\{1\}) = \frac{1}{1 + O_r(12) + O_r(\emptyset) + O_r(2)}, \qquad \Pi_r(\{1,2\}) = \frac{O_r(12)}{1 + O_r(12) + O_r(\emptyset) + O_r(2)},$$

$O_r(12) \to \infty$ gives $\Pi_r(\{1\}) \to 0$ and $\Pi_r(\{1,2\}) \to 1$, and $O_r(12) \to 0$ gives $\Pi_r(\{1\}) \to 1$ and $\Pi_r(\{1,2\}) \to 0$.

The two cases. By Theorem 3.2, $\Pi(\cdot \mid y) = \Pi_{b_*}$ because $s_y^2 = s^2$, and $b_- < b_* < b_+$. For the first claim choose $\gamma \in \big(\frac{2b_- + s^2}{2}, \frac{2b_* + s^2}{2}\big)$. Then $O_{b_-}(12) \to \infty$ and $O_{b_*}(12) \to 0$, so $\Pi_{b_-}(\{1\}) \to 0$ and $\Pi(\{1\} \mid y) \to 1$. For the second choose $\gamma \in \big(\frac{2b_* + s^2}{2}, \frac{2b_+ + s^2}{2}\big)$. Then $O_{b_+}(12) \to 0$ and $O_{b_*}(12) \to \infty$, so $\Pi_{b_+}(\{1,2\}) \to 0$ and $\Pi(\{1,2\} \mid y) \to 1$. For the consequence: if $\Pi(E \mid y) \leq C\,\Pi_{b_-}(E)$ held for all inputs in $\mathcal{B}_n$, the first family with $E = \{1\}$ would give $1 \leq \liminf_n C\,\Pi_{b_-}(\{1\}) = 0$; the second family excludes $b = b_+$ in the same way. □

Remark 6.4. Theorems 6.2 and 6.3 show that a bound on the same-rate posterior $\Pi_{b_-}(E)$ alone, or on $\Pi_{b_+}(E)$ alone, implies no bound on $\Pi(E \mid y)$ on $\mathcal{B}_n$ by a constant factor valid for all inputs. A bound on the mixed quantity of Theorem 6.1, with the numerator at b_- and the denominator at b_+, does imply one, at cost κ^2. Both negative statements are worst cases over inputs. Informally (this is not used), the convergence in Theorem 6.3 is slow: by its proof the log-odds behave like $c \log n$ with $|c| \leq |\gamma/(2r+s^2) - \frac{1}{2}|$, which is small when the rates are small compared with s^2; the statement is qualitative.

7 T02(i): dimension control

Throughout Sections 7–8, G and R hold, f_0 is the $\mathbb{P}_0$-density of y_c, namely $f_0(x) = \mathcal{N}_{n-1}(x; X_c\theta^*, \sigma_*^2 I)$ (Theorem 5.1), and $f_0 > 0$ everywhere. P_S denotes the orthogonal projection in $\mathbb{R}^{n-1}$ onto the column span of $X_{c,S}$, and $\mathsf{M}_S^c = nX_{c,S}H_S^{-1}X_{c,S}^\top$, so $Q_S^c = y_c^\top(I - \mathsf{M}_S^c)y_c$ (Lemma 2.3(a)). Since θ^* is supported on S_0, $X_c\theta^*$ lies in the column span of $X_{c,S}$ for every $S \supseteq S_0$. Also $\|X_c\theta^*\| = \|U^\top\mu\| = \|\mu\|$, because μ lies in the range of $C = UU^\top$.

Lemma 7.1 (numerator)**.** *Fix $b > 0$ not depending on the data. For every $S \in \Omega$, $m_b^c(S)$, as a function of y_c, is a probability density on $\mathbb{R}^{n-1}$. For every set $\mathcal{A} \subseteq \Omega$, $\mathbb{E}_0\big[\sum_{S\in\mathcal{A}} \varpi(S) m_b^c(S)/f_0(y_c)\big] = \varpi(\mathcal{A})$.*

Proof. $m_b^c(S)(x) = \iint g_{S,b}(\theta, v)\, d\theta\, dv$ with $y_c = x$, and $g_{S,b}$ is the joint density of (x, θ, v) under a proper prior. By Tonelli, $\int m_b^c(S)(x)dx = 1$. Since $f_0 > 0$ is the density of y_c, $\mathbb{E}_0[m_b^c(S)(y_c)/f_0(y_c)] = \int m_b^c(S)(x)dx = 1$. Sum over $\mathcal{A}$. □

Lemma 7.2 (denominator)**.** *Assume G, R and $s_0 \le K$, and let b_+ be as in §1. Define $E' = \|\varepsilon_c\|^2/\sigma_*^2 \sim \chi^2_{n-1}$, $\gamma_n = \frac{1}{n+1}\big(1 + \frac{1}{(n-1)\varphi_0^2}\big)$, $R_n = \gamma_n\|P_{S_0}y_c\|^2$ and*

$$\Lambda_n = \frac{|E' - (n-1)|}{2\sqrt{n}} + \frac{E'}{2n} + \frac{R_n}{2\sigma_*^2}.$$

Then, with $c_{\mathrm{IG}} := \min_{v\in[\sigma_*^2, 2\sigma_*^2]} \mathrm{IG}(v; 0.51, b_+) > 0$ (a constant),

$$\frac{\varpi(S_0)m_{b_+}^c(S_0)}{f_0(y_c)} \;\ge\; \tfrac{1}{2}(2p)^{-s_0}(n+1)^{-s_0/2} c_{\mathrm{IG}}\, \sigma_*^2\, n^{-1/2} e^{-\Lambda_n},$$

$$\mathbb{E}_0\Lambda_n \le C_\Lambda := \tfrac{1}{\sqrt{2}} + \tfrac{1}{2} + \mathbf{1}\{s_0 \ge 1\}\frac{(1 + \frac{1}{2\varphi_0^2})(M + \sigma_*^2)}{2\sigma_*^2}.$$

(If $s_0 = 0$, then $R_n = 0$, γ_n is not needed, and φ_0 does not enter.)

Proof. Bound on $Q_{S_0}^c$. If $s_0 \ge 1$, R gives $\lambda_{\min}(G_{S_0}) = (n-1)\lambda_{\min}(R_{S_0}) \ge (n-1)\varphi_0^2$. Lemma 2.3(c) with $g = (n-1)\varphi_0^2$ gives $I - \mathsf{M}_{S_0}^c \preceq (I - P_{S_0}) + \gamma_n P_{S_0}$, so $Q_{S_0}^c \le \|(I-P_{S_0})y_c\|^2 + R_n = \|(I-P_{S_0})\varepsilon_c\|^2 + R_n \le \|\varepsilon_c\|^2 + R_n$, because $(I - P_{S_0})X_c\theta^* = 0$. If $s_0 = 0$, then $\mu = 0$ and $Q_\emptyset^c = \|y_c\|^2 = \|\varepsilon_c\|^2$. In both cases $Q_{S_0}^c \le \sigma_*^2 E' + R_n$.

Integral representation. By Lemma 2.1 and Tonelli,

$$m_{b_+}^c(S_0) = \int_0^\infty (2\pi v)^{-(n-1)/2} D_{S_0}^{-1/2} e^{-Q_{S_0}^c/(2v)} \mathrm{IG}(v; 0.51, b_+)\, dv.$$

Since $f_0(y_c) = (2\pi\sigma_*^2)^{-(n-1)/2}e^{-E'/2}$,

$$\frac{m_{b_+}^c(S_0)}{f_0(y_c)} = D_{S_0}^{-1/2}\int_0^\infty \exp\Big\{-\frac{n-1}{2}\log\frac{v}{\sigma_*^2} + \frac{E'}{2} - \frac{Q_{S_0}^c}{2v}\Big\}\mathrm{IG}(v; 0.51, b_+)\, dv.$$

Lower bound on the exponent. Let $v = \sigma_*^2 u$ with $1 \le u \le 1 + n^{-1/2}$. Using $\log u \le u - 1$, $Q_{S_0}^c \le \sigma_*^2 E' + R_n$ and $1/u \ge 2 - u$,

$$-\frac{n-1}{2}\log u + \frac{E'}{2} - \frac{Q_{S_0}^c}{2\sigma_*^2 u} \ge \frac{u-1}{2}\Big(\frac{E'}{u} - (n-1)\Big) - \frac{R_n}{2\sigma_*^2} \ge \frac{u-1}{2}\big(E' - (n-1) - E'(u-1)\big) - \frac{R_n}{2\sigma_*^2}$$
$$\ge -\frac{|E' - (n-1)|}{2\sqrt{n}} - \frac{E'}{2n} - \frac{R_n}{2\sigma_*^2} = -\Lambda_n.$$

Here the first step uses $-\frac{n-1}{2}(u-1)+\frac{E'}{2}(1-\frac{1}{u})=\frac{u-1}{2}(\frac{E'}{u}-(n-1))$, and the last uses $0\le u-1\le n^{-1/2}$.

Assembling. Restrict the integral to $v\in[\sigma_*^2,\sigma_*^2(1+n^{-1/2})]\subseteq[\sigma_*^2,2\sigma_*^2]$, an interval of length $\sigma_*^2 n^{-1/2}$ on which $\mathrm{IG}(v;0.51,b_+)\ge c_{\mathrm{IG}}$. Use $D_{S_0}\le(n+1)^{s_0}$ (Lemma 2.3(e)) and $\varpi(S_0)\ge\frac{1}{2}(2p)^{-s_0}$ (Lemma 2.4(d)). The constant c_{IG} is the minimum of a continuous positive function on a compact interval, and it depends only on (σ_*^2,M).

Mean of Λ_n. By Lemma 2.5(b), $\mathbb{E}|E'-(n-1)|\le\sqrt{2(n-1)}\le\sqrt{2n}$, and $\mathbb{E}E'=n-1$. Also $\mathbb{E}R_n\le\gamma_n\mathbb{E}\|y_c\|^2=\gamma_n(\|\mu\|^2+(n-1)\sigma_*^2)\le\gamma_n n(M+\sigma_*^2)\le(1+\frac{1}{2\varphi_0^2})(M+\sigma_*^2)$, using $n/(n+1)\le1$ and $n-1\ge2$. □

Theorem 7.3 (T02(i))**.** *Assume G and R (regime R1). For every n and every integer m define $\Theta_{n,m}\ge0$ by $\Pi(|S|>m\mid y)=2^{-m}(2p)^{s_0}(n+1)^{s_0/2}\sqrt{n}\,\Theta_{n,m}$. Then $\lim_{L\to\infty}\limsup_{n\to\infty}\sup_m\mathbb{P}_0(\Theta_{n,m}>L)=0$. Consequently $\Pi(|S|>m_n(\omega_n)\mid y)=O_{\mathbb{P}_0}(e^{-\omega_n})$ for every deterministic sequence ω_n.*

Proof. Let n be large enough that $s_0\le K$ (R). Put $E_m=\{|S|>m\}$ and $N_m=I_{b_-}(E_m)/f_0(y_c)=\sum_{S\in\Omega,|S|>m}\varpi(S)m^c_{b_-}(S)/f_0(y_c)$. The rate b_- does not depend on the data, so Lemma 7.1 and Lemma 2.4(c) give $\mathbb{E}_0N_m=\varpi(|S|>m)\le2^{1-m}$. Put $\Theta^{(1)}_{n,m}=N_m/2^{1-m}$; then $\mathbb{E}_0\Theta^{(1)}_{n,m}\le1$ for every integer m. On $\mathcal{B}_n$, Theorem 6.1 and Lemma 7.2 give

$$\Pi(|S|>m\mid y)\le\kappa^2\frac{N_m}{\varpi(S_0)m^c_{b_+}(S_0)/f_0(y_c)}\le\kappa^2\,2^{1-m}\Theta^{(1)}_{n,m}\cdot\frac{2(2p)^{s_0}(n+1)^{s_0/2}\sqrt{n}\,e^{\Lambda_n}}{c_{\mathrm{IG}}\sigma_*^2}.$$

That is, on $\mathcal{B}_n$, $\Theta_{n,m}\le c_1e^{\Lambda_n}\Theta^{(1)}_{n,m}$ with $c_1=4\kappa^2/(c_{\mathrm{IG}}\sigma_*^2)$, a constant. Hence, for $L>c_1^2$,

$$\mathbb{P}_0(\Theta_{n,m}>L)\le\mathbb{P}_0(\mathcal{B}_n^c)+\mathbb{P}_0(\Theta^{(1)}_{n,m}>\sqrt{L})+\mathbb{P}_0\big(c_1e^{\Lambda_n}>\sqrt{L}\big)\le\mathbb{P}_0(\mathcal{B}_n^c)+\frac{1}{\sqrt{L}}+\frac{C_\Lambda}{\log(\sqrt{L}/c_1)},$$

by Markov's inequality applied to $\Theta^{(1)}_{n,m}$ and to $\Lambda_n\ge0$. The last two terms do not depend on n or m. By Theorem 5.1, $\mathbb{P}_0(\mathcal{B}_n^c)\to0$. So $\limsup_n\sup_m\mathbb{P}_0(\Theta_{n,m}>L)\le L^{-1/2}+C_\Lambda/\log(\sqrt{L}/c_1)\to0$ as $L\to\infty$.

For the consequence, let $m=m_n(\omega_n)$. By definition, $m\log2\ge s_0\log(2p)+\frac{s_0+1}{2}\log n+\omega_n$, i.e. $2^{-m}\le(2p)^{-s_0}n^{-(s_0+1)/2}e^{-\omega_n}$. So $\Pi(|S|>m\mid y)\le(1+1/n)^{s_0/2}e^{-\omega_n}\Theta_{n,m}\le2^{s_0/2}e^{-\omega_n}\Theta_{n,m}$. Moreover $\mathbb{P}_0(\Theta_{n,m_n(\omega_n)}>L)\le\sup_m\mathbb{P}_0(\Theta_{n,m}>L)$, which gives tightness of $e^{\omega_n}\Pi(|S|>m_n(\omega_n)\mid y)$. □

Remark 7.4. The proof uses from R only $\|\mu\|^2\le nM$, $\lambda_{\min}(R_{S_0})\ge\varphi_0^2$ and eventually $s_0\le K$. It uses neither $p\le n^A$ nor any eigenvalue condition on models other than S_0, and it holds for every deterministic ω_n; the restrictions $\omega_n\to\infty$, $\omega_n=O(\log n)$ in `STATEMENT.md` are not needed. The mixed-endpoint form of Theorem 6.1 is essential here. The numerator identity of Lemma 7.1 is proved only for a rate that does not depend on the data; for the data-dependent rate $b_0s_y^2$ the function $y_c\mapsto m^c_{b_0s_y^2(y_c)}(S)$ need not integrate to 1, and this remark (not used anywhere) makes no claim about it. The denominator bound needs a fixed rate so that c_{IG} is a constant.

8 T02(ii): strict supersets

In this section G, R and SE hold, and $m_n=m_n(\frac{1}{2}\log n)$. Since $p\le n^A$ and $\log n\ge1$ for $n\ge3$,

$$m_n\le1+\frac{s_0(\log2+A\log n)+\frac{s_0+2}{2}\log n}{\log2}\le C_m\log n,\qquad C_m:=1+\frac{s_0(1+A)+\frac{s_0+2}{2}}{\log2}.\tag{5}$$

Put $C_A = 10+4A$, $\tau_j = C_A j \log n$, $\eta_n = \frac{1}{(n+1)(n-1)\varphi_0^2}$ and $\delta_n = \frac{\log n}{\sqrt{n-1}} + \frac{C_A m_n \log n}{n-1}$. Then $\delta_n \log n \to 0$ by (5). For $J \subseteq [p] \setminus S_0$ with $J \neq \emptyset$ and $s_0 + |J| \le m_n$, SE applies to $S_0 \cup J$: it is nonempty and $|S_0 \cup J| \le \min(p, m_n) \le \min(p, m_n + s_0)$, because $S_0 \cup J \subseteq [p]$. So $X_{c,S_0\cup J}$ has full column rank, $\lambda_{\min}(G_{S_0\cup J}) \ge (n-1)\varphi_0^2$, and $P_{S_0\cup J} - P_{S_0}$ is the orthogonal projection onto the orthogonal complement of $\operatorname{span}(X_{c,S_0})$ in $\operatorname{span}(X_{c,S_0\cup J})$, of rank $|J|$. Define

$$\chi_J^2 := \|(P_{S_0\cup J} - P_{S_0})\varepsilon_c\|^2/\sigma_*^2 \;\sim\; \chi^2_{|J|} \quad \text{exactly.}$$

Definition (good event)**.** *$\mathcal{G}_n$ is the intersection of*

(G1) $|\|\varepsilon_c\|^2 - (n-1)\sigma_*^2| \le \sigma_*^2\sqrt{n-1}\log n$*;*

(G2) $\|P_T\varepsilon_c\|^2 \le \sigma_*^2 C_A m_n \log n$ *for every* $T \subseteq [p]$ *with* $1 \le |T| \le m_n$*;*

(G3) $\chi_J^2 \le \tau_{|J|}$ *for every* $J \subseteq [p] \setminus S_0$ *with* $J \neq \emptyset$ *and* $s_0 + |J| \le m_n$*.*

Lemma 8.1. *Under G and* $p \le n^A$*:* $\mathbb{P}_0(\mathcal{G}_n^c) \le 2/\log^2 n + 2/n$*.*

Proof. (G1): $\|\varepsilon_c\|^2/\sigma_*^2 \sim \chi^2_{n-1}$, and Lemma 2.5(b) with $t = \sqrt{n-1}\log n$ gives probability at most $2/\log^2 n$. For (G2) and (G3) let $x_k = k(\log p + 1) + \log n$. For $k \ge 1$, using $\log p \le A \log n$ and $1 \le \log n$,

$$2k\log 2 + 4x_k \le k\big(2\log 2 + 4A\log n + 4\big) + 4\log n \le (10+4A)k\log n = C_A k \log n.$$

(G2): for $|T| = k$, $\|P_T\varepsilon_c\|^2/\sigma_*^2 \sim \chi^2_{\operatorname{rank}(X_{c,T})}$ with rank $\le k$. By Lemma 2.5(a), $\mathbb{P}(\|P_T\varepsilon_c\|^2/\sigma_*^2 > C_A k \log n) \le e^{-x_k}$. There are at most p^k sets T of size k, so the union over $k \ge 1$ has probability at most $\sum_k p^k e^{-x_k} = \sum_{k\ge1} e^{-k}/n \le 1/n$. On the complement, $\|P_T\varepsilon_c\|^2 \le \sigma_*^2 C_A |T| \log n \le \sigma_*^2 C_A m_n \log n$. (G3): the same computation with the at most p^j sets J of size j and threshold $\tau_j = C_A j \log n$ gives probability at most $1/n$. □

Lemma 8.2 (on $\mathcal{G}_n$)**.** *There is* n_0 *(depending only on* $s_0, A, \varphi_0, \sigma_*^2, M$*) such that, for* $n \ge n_0$*,* $s_0 \le K$ *and, on* $\mathcal{G}_n$*:*

(a) $\|y_c\|^2 \le 4n(M + \sigma_*^2)$*;*

(b) for every $S \in \Omega$ *with* $S \supseteq S_0$ *and* $|S| \le m_n$*:* $Q_S^c \ge (n-1)\sigma_*^2(1-\delta_n)$*, and* $\delta_n \le \frac{1}{2}$*;*

(c) for every $S = S_0 \cup J \in \Omega$ *with* $J \subseteq [p] \setminus S_0$*,* $J \neq \emptyset$*,* $|S| \le m_n$*:* $0 \le Q_{S_0}^c - Q_S^c \le \frac{n}{n+1}\big[\sigma_*^2\chi_J^2 + \eta_n\|y_c\|^2\big]$*.*

Proof. Take n_0 with $\log n \le \sqrt{n-1}$, $\delta_n \le \frac{1}{2}$ and $s_0 \le K$ for $n \ge n_0$; this is possible because $\delta_n \to 0$ and because $s_0 \le p$ (as $S_0 \subseteq [p]$) gives $s_0 \le K$ once $n \ge 2s_0 + 1$. (a) By (G1), $\|\varepsilon_c\|^2 \le 2(n-1)\sigma_*^2$; so $\|y_c\|^2 \le 2\|X_c\theta^*\|^2 + 2\|\varepsilon_c\|^2 \le 2nM + 4(n-1)\sigma_*^2$. (b) By Lemma 2.3(b), $Q_S^c \ge \|(I - P_S)y_c\|^2 = \|(I-P_S)\varepsilon_c\|^2 = \|\varepsilon_c\|^2 - \|P_S\varepsilon_c\|^2$, because $X_c\theta^* \in \operatorname{span}(X_{c,S})$. Then (G1) and (G2) apply (if $S = \emptyset$, $P_S = 0$). (c) $Q_{S_0}^c - Q_S^c = y_c^\top(\mathsf{M}_S^c - \mathsf{M}_{S_0}^c)y_c \ge 0$ by Lemma 2.3(d). By Lemma 2.3(b), $\mathsf{M}_S^c \preceq \frac{n}{n+1}P_S$. If $s_0 \ge 1$, Lemma 2.3(c) with $g = (n-1)\varphi_0^2$ (R) gives $\mathsf{M}_{S_0}^c \succeq \frac{n}{n+1}(1-\eta_n)P_{S_0}$; if $s_0 = 0$ both sides vanish. So $\mathsf{M}_S^c - \mathsf{M}_{S_0}^c \preceq \frac{n}{n+1}\big[(P_S - P_{S_0}) + \eta_n P_{S_0}\big]$. Both projections fix $X_c\theta^*$, so $(P_S - P_{S_0})y_c = (P_S - P_{S_0})\varepsilon_c$ and $\|(P_S - P_{S_0})\varepsilon_c\|^2 = \sigma_*^2\chi_J^2$. Finally $\|P_{S_0}y_c\|^2 \le \|y_c\|^2$. □

Lemma 8.3 (one superset)**.** *For $n \ge n_0$, on $\mathcal{G}_n$, for every $S = S_0 \cup J$ as in Lemma 8.2(c) with $j = |J|$, and every $b > 0$,*

$$\frac{\varpi(S)\, m_b^c(S)}{\varpi(S_0)\, m_b^c(S_0)} \le \frac{w(s_0+j)}{w(s_0)} (n+1)^{-j/2} \Big(1 + \frac{1}{(n-1)\varphi_0^2}\Big)^{j/2} e^{j\zeta_n}\, e^{\chi_J^2/2}\, e^{\rho_n},$$

where $\vartheta_n = \frac{n(n+0.02)}{(n^2-1)(1-\delta_n)}$, $\zeta_n = \frac{1}{2}(\vartheta_n - 1) C_A \log n$ and $\rho_n = \frac{2n^2(n+0.02)(M+\sigma_*^2)\eta_n}{(n+1)(n-1)\sigma_*^2(1-\delta_n)}$. Moreover $\zeta_n \to 0$ and $\rho_n \to 0$.

Proof. By (3), $\frac{m_b^c(S)}{m_b^c(S_0)} = \big(\frac{D_{S_0}}{D_S}\big)^{1/2} \big(\frac{2b+Q_{S_0}^c}{2b+Q_S^c}\big)^{\nu}$. Here $Q_{S_0}^c \ge Q_S^c > 0$ by Lemma 8.2(b),(c). For $x \ge y > 0$ the map $b \mapsto (2b+x)/(2b+y)$ is nonincreasing on $b \ge 0$, so its value at $b > 0$ is at most its value x/y at $b = 0$. So the second factor is at most $(Q_{S_0}^c/Q_S^c)^\nu = (1+\xi)^\nu \le e^{\nu\xi}$, with $\xi = (Q_{S_0}^c - Q_S^c)/Q_S^c \ge 0$. By Lemma 8.2(a)–(c) and $\nu = (n+0.02)/2$,

$$\nu\xi \le \frac{(n+0.02)\, n\, [\sigma_*^2 \chi_J^2 + \eta_n 4n(M+\sigma_*^2)]}{2(n+1)(n-1)\sigma_*^2(1-\delta_n)} = \frac{\vartheta_n}{2}\chi_J^2 + \rho_n.$$

On $\mathcal{G}_n$, (G3) gives $\chi_J^2 \le \tau_j$. Since $\vartheta_n \ge 1$ (as $n(n+0.02) \ge n^2 - 1$), $\frac{\vartheta_n}{2}\chi_J^2 \le \frac{1}{2}\chi_J^2 + \frac{1}{2}(\vartheta_n - 1)\tau_j = \frac{1}{2}\chi_J^2 + j\zeta_n$. For the determinant, SE gives $\lambda_{\min}(G_S) \ge (n-1)\varphi_0^2$. Lemma 2.3(e) with $g = (n-1)\varphi_0^2$ then gives $(D_{S_0}/D_S)^{1/2} \le (n+1)^{-j/2}(1 + 1/((n-1)\varphi_0^2))^{j/2}$. The prior ratio is $\varpi(S)/\varpi(S_0) = w(s_0+j)/w(s_0)$, since both models are in Ω. Finally, $\vartheta_n - 1 = \frac{0.02n+1+(n^2-1)\delta_n}{(n^2-1)(1-\delta_n)} \le 2\big(\delta_n + \frac{0.02n+1}{n^2-1}\big)$, so $\zeta_n \le C_A \log n\, (\delta_n + \frac{0.02n+1}{n^2-1}) \to 0$; and $\rho_n = O(1/n)$ because $\eta_n = O(n^{-2})$. □

Theorem 8.4 (T02(ii))**.** *Under G, R and SE in regime R1, $\Pi(\{S \in \Omega : S \supsetneq S_0\} \mid y) = O_{\mathbb{P}_0}(r_n)$ with $r_n = \sqrt{\log n / n}$.*

Proof. Let $n \ge n_0$. Split $\Pi(S \supsetneq S_0 \mid y) \le \Pi(S \supsetneq S_0, |S| \le m_n \mid y) + \Pi(|S| > m_n \mid y) =: T_1 + T_2$.

The term T_1. By Theorem 3.2, $\pi(S)/\pi(S_0) = \varpi(S) m_b^c(S)/(\varpi(S_0) m_b^c(S_0))$ with $b = b_0 s_y^2$, and $\pi(S_0) \le 1$. By Lemma 8.3, on $\mathcal{G}_n$,

$$T_1 \le \sum_{S \supsetneq S_0,\, |S| \le m_n} \frac{\pi(S)}{\pi(S_0)} \le e^{\rho_n} V_n, \qquad V_n := \sum_{j \ge 1} \sum_{\substack{J \subseteq [p] \setminus S_0,\ |J| = j \\ s_0 + j \le \min(m_n, K)}} \frac{w(s_0+j)}{w(s_0)} a_n^j\, e^{\chi_J^2/2} \mathbf{1}\{\chi_J^2 \le \tau_j\},$$

where $a_n = (n+1)^{-1/2}(1 + \frac{1}{(n-1)\varphi_0^2})^{1/2} e^{\zeta_n}$. The indicator equals 1 on $\mathcal{G}_n$ by (G3). V_n is a sum over a deterministic index set of nonnegative variables, and each χ_J^2 in it is exactly χ_j^2. By Lemma 2.5(c), $\mathbb{E}_0[e^{\chi_J^2/2} \mathbf{1}\{\chi_J^2 \le \tau_j\}] \le (e\tau_j/j)^{j/2} = (eC_A \log n)^{j/2}$. By Lemma 2.4(e) (applicable since $s_0 + j \le K \le p$),

$$\mathbb{E}_0 V_n \le \sum_{j \ge 1} 2(j+1)^{s_0} 2^{-j} a_n^j (eC_A \log n)^{j/2} = 2 \sum_{j \ge 1} (j+1)^{s_0} \bar{r}_n^j, \qquad \bar{r}_n := \tfrac{1}{2} a_n \sqrt{eC_A \log n}.$$

Since $\zeta_n \to 0$, $\bar{r}_n \le c_2 r_n$ for a constant c_2 and all large n, and $\bar{r}_n \le \frac{1}{2}$ eventually. Then $\mathbb{E}_0 V_n \le 2\bar{r}_n \sum_{j \ge 1} (j+1)^{s_0} 2^{-(j-1)} = 2K_{s_0} \bar{r}_n$ with $K_{s_0} < \infty$. Hence, by Markov's inequality, for every $L > 0$ and all large n,

$$\mathbb{P}_0(T_1 > L r_n) \le \mathbb{P}_0(\mathcal{G}_n^c) + \mathbb{P}_0(e^{\rho_n} V_n > L r_n) \le \frac{2}{\log^2 n} + \frac{2}{n} + \frac{2e^{\rho_n} K_{s_0} c_2}{L}.$$

The term T_2. Theorem 7.3 with $\omega_n = \frac{1}{2}\log n$ gives $T_2 \le 2^{s_0/2} n^{-1/2} \Theta_{n,m_n} \le 2^{s_0/2} r_n \Theta_{n,m_n}$ (as $\log n \ge 1$), with $\lim_L \limsup_n \mathbb{P}_0(\Theta_{n,m_n} > L) = 0$.

Combining, $\limsup_n \mathbb{P}_0\big(\Pi(S \supsetneq S_0 \mid y) > 2Lr_n\big) \le \frac{2K_{s_0}c_2}{L} + \limsup_n \mathbb{P}_0(\Theta_{n,m_n} > 2^{-s_0/2} L) \to 0$ as $L \to \infty$. □

Remark 8.5. T02(ii) needs no rate envelope. For supersets of S_0 the v3 Bayes factor against S_0 is nonincreasing in the IG rate, so it is bounded at the data-dependent rate $b_0 s_y^2$ by its value at rate 0 (Lemma 8.3). The only use of T01 is the identity (i), and the scale of s_y never enters the bound. The envelope enters only through the tail T_2, via T02(i).